

EXPERIMENTAL INVESTIGATION OF THE CRITICAL IONIZATION VELOCITY IN GAS MIXTURES*

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Abstract. The critical ionization velocity which is of cosmogonic and astrophysical interest has hitherto mainly been investigated for pure gases. Since in space we always have gas mixtures, it is of interest also to study gas mixtures. The present report, which is a summary of a more detailed report (Axnäs, 1976), summarizes the results of systematic experiments on the critical ionization velocity as a function of the mixing ratio for binary gas mixtures of H_2 , He, N_2 , O_2 , Ne and Ar. The apparatus used is a coaxial plasma gun with an azimuthal magnetic field. The discharge parameters are chosen so that the plasma is weakly ionized. In some of the mixtures it is found that one of the components tends to dominate in the sense that only a small amount (regarding volume) of that component is needed for the discharge to adopt a limiting velocity close to that for the pure component. Thus in a mixture between a heavy and a light component having nearly equal ionization potentials, the heavy component dominates. Also, if there is a considerable difference in ionization potential between the components, the component with the lowest ionization potential tends to dominate.

1. Introduction

In his theory for the origin of the solar system Alfvén (Alfvén, 1954; Alfvén and Wilcox, 1962; Alfvén and Arrhenius, 1976) made an assumption that a strong interaction will occur between a magnetized plasma and a neutral gas when the neutral particles in the coordinate system of the plasma reach a velocity so that the kinetic energy equals the ionization energy

$$u_c = \left(\frac{2eV_i}{m_n} \right)^{1/2}, \quad (1)$$

where e is the electron charge, V_i the ionization potential, and m_n the neutral particle mass. By use of this hypothesis the mass distribution of the planetary system and satellite systems of the giant planets could be predicted fairly well. Later the phenomenon has been discovered in different crossed field experiments – in the beginning accidentally, later through systematic studies. The critical ionization velocity, as it seems from experiments, is remarkably independent of experimental parameters such as discharge current (plasma density), neutral gas density, magnetic field and experimental geometry. This behaviour of the $E \times B$ discharge indicates that regarding particle balance we may have a situation illustrated schematically in Figure 1. The E/B of the point where the two curves intersect each other is comparatively

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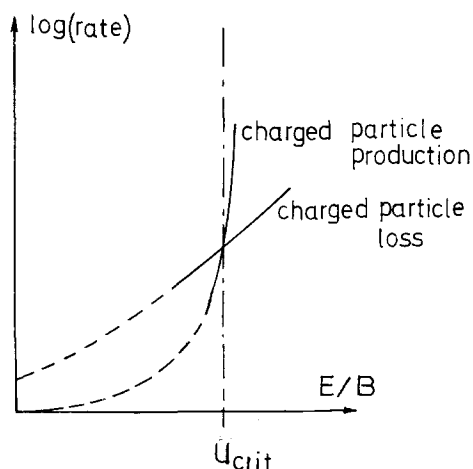


Fig. 1. A tentative illustration of the particle balance in a critical ionization velocity situation.

independent of the inclination of the loss curve as long as the inclination of the production curve is still higher. The loss curve could be strongly dependent on the discharge parameters, but still the E/B value will not change very much.

In space the critical ionization velocity phenomenon may be of importance. Thus it has been considered as a possible explanation of the observed anomalous ionization of gas released from the lunar surface (Freeman *et al.*, 1972). Values of the critical velocity for different gases are shown in Table I. Review articles on the critical ionization velocity phenomenon are given by Danielsson (1973) and Sherman (1973).

It seems, at least from the experimental point of view, that each gas has its own characteristic critical ionization velocity. Since in space we always have mixtures of gases, it is then of general interest to determine the critical ionization velocity for gas mixtures. Of direct interest is such an investigation in connection with the cosmogonic theory (Alfvén and Arrhenius, 1976; Arrhenius, 1976). The present report, which is a summary of a more detailed report (Axnäs, 1976), summarizes the results

TABLE I

Gas	Atomic weight	Ionization potential (volts)	Critical ionization velocity u_c ($\text{ms}^{-1} \times 10^{+4}$)	Ion gyro radius at u_c ($\text{m} \times 10^{-3}$) $B = 0.66 \text{ T}$
Hydrogen	1	13.6	5.1	0.8
Helium	4	24.6	3.4	2.2
Nitrogen	14	14.5	1.4	3.1
Oxygen	16	13.6	1.3	3.2
Neon	20	21.6	1.6	4.5
Argon	40	15.8	0.9	5.5

Figs. 2-13. Measured (circles) and calculated (Equation (3), solid line) values of E/B as functions of the filling gas composition expressed in volume per cent. The arrows show where the calculated curves intersect the vertical axis.

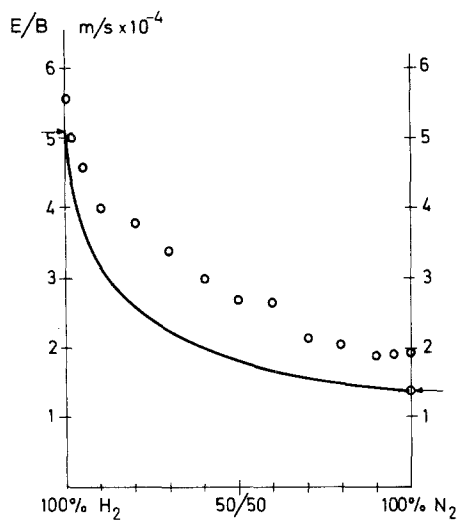


Fig. 2

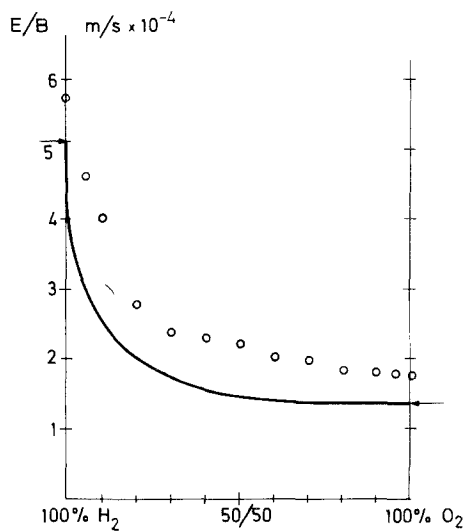


Fig. 3

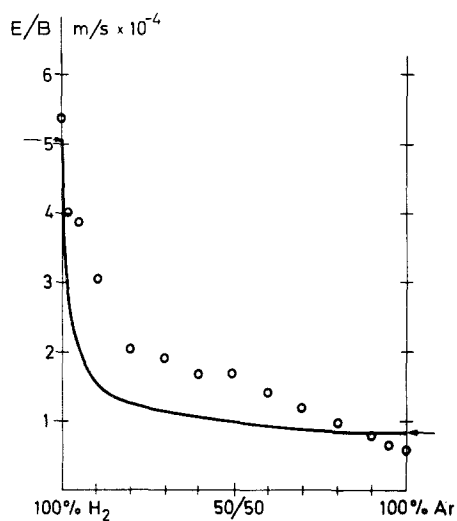


Fig. 4

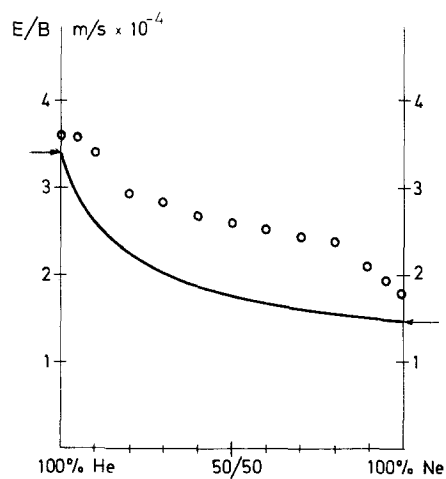


Fig. 5

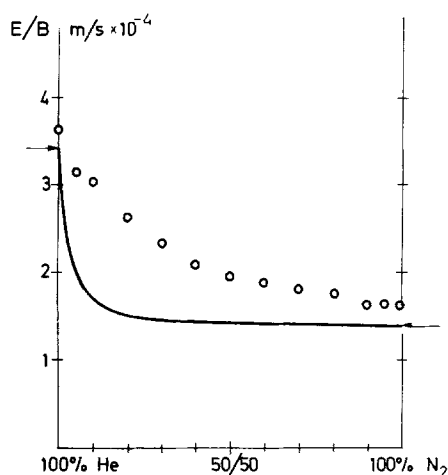


Fig. 6

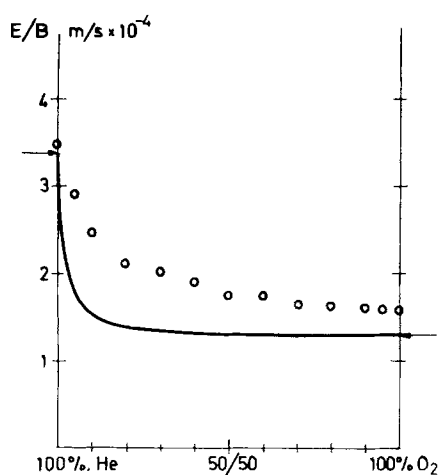


Fig. 7

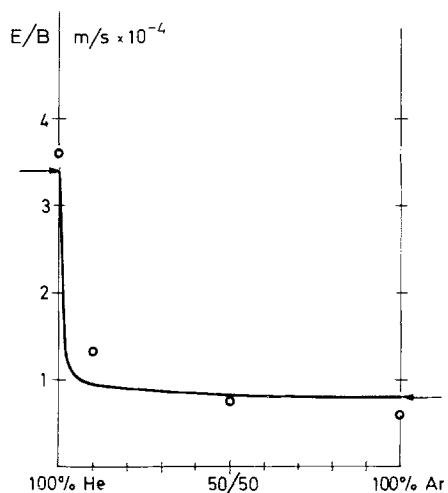


Fig. 8

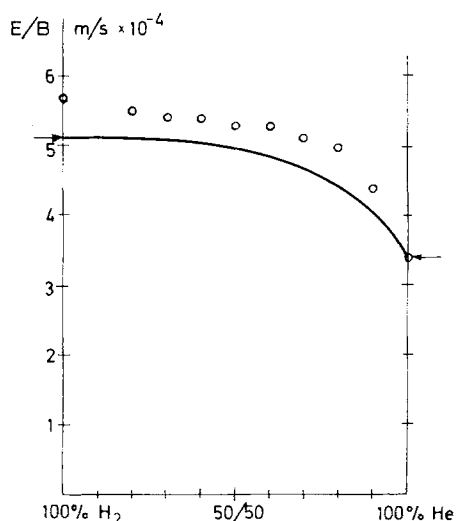


Fig. 9

of systematic experiments on the critical ionization velocity as a function of the mixing ratio for several different binary gas mixtures.

2. Experimental Method and Parameters

The apparatus used in this experiment is a coaxial plasma gun with an azimuthal magnetic field and a radial discharge current. The gas is continuously let into the gun from a mixing tank via a variable leak. The filling pressure (neutral gas density) is adjusted by varying the pumping speed. It was checked that no enrichment of any of the components occurred (Axnäs, 1976).

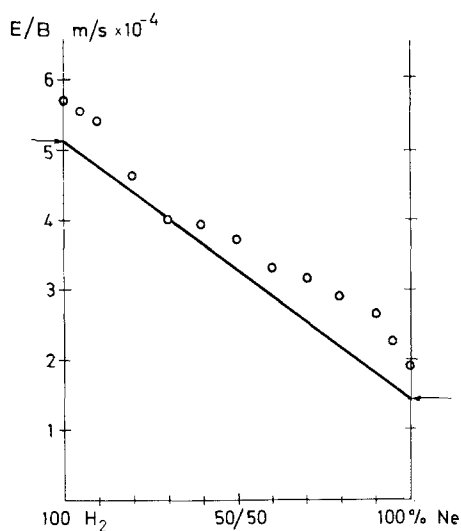


Fig. 10

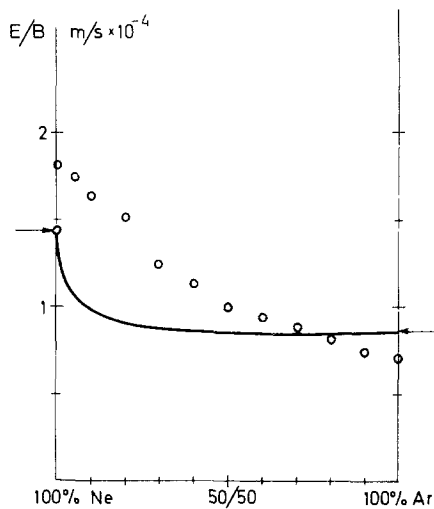


Fig. 11

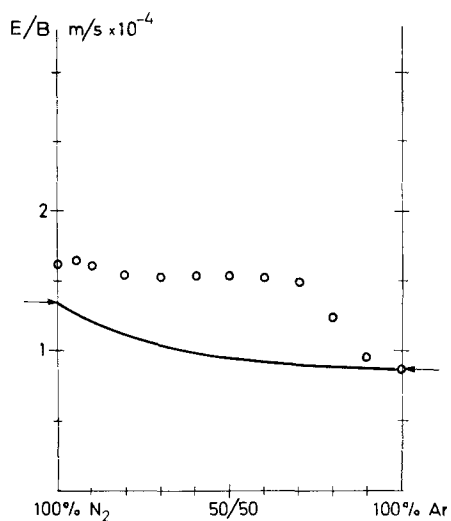


Fig. 12

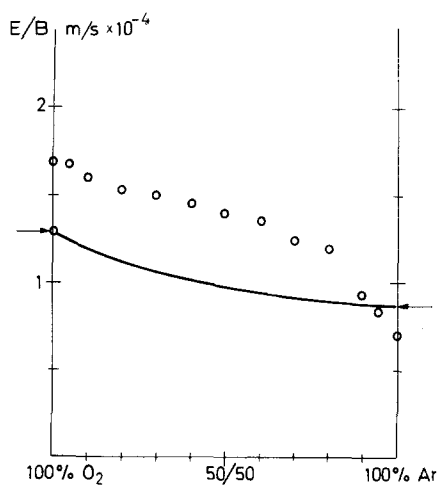


Fig. 13

As a first step it has been found reasonable to choose gas mixtures with components having notable differences in measured E/B (critical ionization velocity). Choosing the discharge current, 100 A, filling pressure, 0.1 torr ($n = 3.3 \times 10^{21} \text{ m}^{-3}$) and the magnetic field, 0.1–1 T, we achieved a weakly ionized gas and good azimuthal symmetry (hydrogen discharge). The gun and experiments performed in pure gases using this have been described in earlier papers (Wilcox *et al.*, 1964; Eninger, 1966; Axnäs, 1972; Raadu, 1977).

3. Results

In Figures 2–13 the experimentally obtained ratio E/B for a gas mixture discharge is shown as a function of the neutral gas composition (expressed in volume per cent). The different gas mixtures could be divided into three groups:

- (1) The components have roughly the same ionization potential but considerably different molecule (atom) weights (H_2 – N_2 , H_2 – O_2 , H_2 –Ar and He–Ne, Figures 2, 3, 4 and 5).
- (2) One component has much larger ionization potential (He– N_2 , He– O_2 , He–Ar, Figures 6, 7 and 8; H_2 –He, H_2 –Ne, Ne–Ar, Figures 9, 10 and 11).
- (3) The difference regarding both molecular weight and ionization potential are not large (N_2 –Ar and O_2 –Ar, Figures 12 and 13).

An experimental result common to all mixtures is that E/B for any composition has a value between those for the pure components.

From Figures 2–4, it is clear that in most cases in the first group the heavier component dominates, i.e. a relatively small amount (regarding volume) of the heavier component brings the E/B of the mixture close to that for the pure heavy component. However, in the He–Ne mixture (Figure 5) none of the components dominates. In the second group the component with the lowest ionization potential dominates in most cases. In the Ne–Ar mixture (Figure 7) this effect is not so pronounced and in the H_2 –Ne mixture (Figure 6) none of the components dominates. In the third group the molecular gases O_2 and N_2 dominate in mixtures with Ar (Figures 9 and 10).

The statistical error of the measured values of E/B is typically $\pm 10\%$. In some cases, when the dependence on the composition is very strong, errors in the partial pressures of the gas in the mixing tank could increase the error to a value of $\pm 15\%$.

4. Discussion

As yet no definite theoretical explanation of the critical velocity phenomenon exists and no complete discussion will be attempted here. However, some features of the experimental results will be discussed.

In the gas mixtures of group 1, by introducing 10% of the heavier component the value of E/B will drop to about the mean value of the E/B for the pure components (not in He–Ne mixtures). The heavy component could influence the discharge in several different ways:

- (1) The Hall and Pedersen conductivities in a weakly ionized plasma may be expressed as functions of the plasma density, the electron and ion Hall parameters (Alfvén and Fälthammar, 1963). If the electric field has both radial and axial components, the radial current density will depend on both the Pedersen and the Hall conductivities. The heavy ions (which may have small values of the Hall parameter) could change the Pedersen and Hall conductivities and, hence, the electric field. Also the current distribution could be changed.

- (2) In some theories for the critical ionization velocity the electrons are heated through energy transfer from the ions (Lehnert, 1967; Sherman, 1972; Raadu, 1975). The introduction of heavy ions, which are able to gain large energy from the electric field in the discharge, could increase such energy transfer. This would give increased ionization (and hence conductivity).
- (3) Processes leading to increased heavy ion abundance (e.g. ion-neutral reactions and other selective loss mechanisms) will amplify both the mechanisms mentioned above.

The tendency of the mixtures belonging to group 2 to adopt the E/B of the component with the lowest ionization potential is somewhat easier to understand. A small amount of the component with low ionization potential will increase the total ionization (and hence the conductivity) appreciably.

Alfvén's original hypothesis is valid for pure gases. This has been verified by a number of experiments (Danielsson, 1973). A natural way of generalizing Alfvén's hypothesis is to equate the kinetic energy supply associated with the formation of ions moving relative to the plasma with the loss rate through ionization (Raadu, 1976) – i.e.

$$\alpha \frac{1}{2} m_{\text{II}} u_c^2 + (1 - \alpha) \frac{1}{2} m_{\text{III}} u_c^2 = \alpha e V_{\text{II}} + (1 - \alpha) e V_{\text{III}}, \quad (2)$$

where $\alpha = \nu_{\text{II}}/(\nu_{\text{II}} + \nu_{\text{III}})$ is the fractional ion production rate of component I.

This is also in basic agreement with the modified two-stream instability (Sherman, 1972; Raadu, 1975) due to which the energy source driving the instability is the relative motion between the electrons and the newly formed ions. The electron temperature used to calculate α is chosen so that the total electron ionization frequency $\nu_{\text{II}} + \nu_{\text{III}}$ is equal to the one found experimentally, 10^6 s^{-1} (Axnäs, 1972). E/B calculated from Equation (2) is shown in Figures 2–13 (solid curves). In molecular gases the situation is more complicated. First, molecular as well as atomic ions could be formed. Second, in the case of dissociative ionization the dissociation energy has to be taken into account. However, for molecular gases Equation (2) is shown in Figures 2–13 only for the case with atomic mass (no dissociation energy). From Figures 2–13 it is clear that in almost all cases (except in argon-dominated mixtures) the experimental values are somewhat larger than those calculated. However, there is a basic similarity regarding composition dependence between the calculated and the measured values of E/B .

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